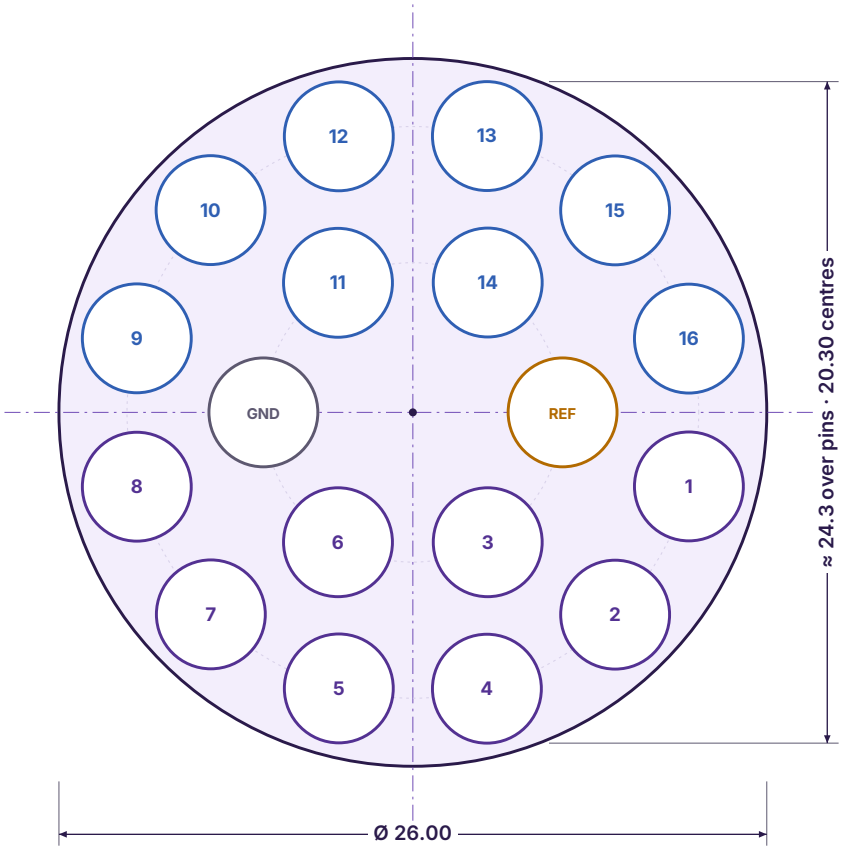
PIEEG

PIN-SIDE VIEW · +y toward the stacking connector

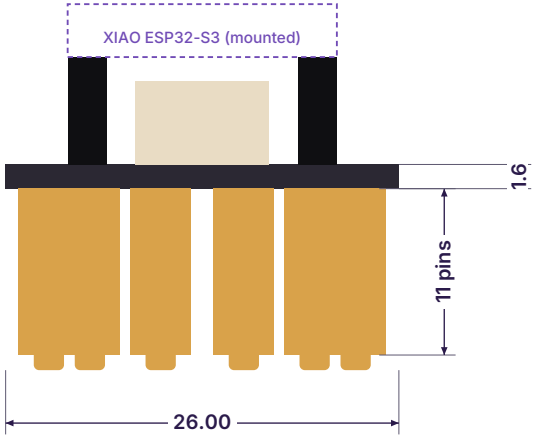


Pitch ≈ 5.38 (hexagonal)

Inner ring r ≈ 5.5: REF, GND, ch 3, 6, 11, 14

Outer ring r ≈ 10.5: remaining 12 channels

SIDE ELEVATION (schematic)



Pogo travel compresses under strap pressure; socket and XIAO heights depend on the parts fitted and are shown for orientation only.

Notes

- Round board Ø 26.00, 1.6 mm, 4 layers.
- 18 pogo pins Ø4 × 11 mm on a hexagonal ≈ 5.38 mm grid.
- REF at +5.50, 0 and GND at −5.49, 0 (inner ring).
- XIAO sockets on the component side, rows 15.24 mm apart.
- No mounting holes: the board is held by the XIAO and a holder or strap.

Title	Octopus 16 outline & electrode array
Doc. No.	OC16-MD-002
Revision	Rev. A · 10 Nov 2026
Units	mm; datum = board centre
View	Pin side, +y toward the stacking connector
Source	Production layout data

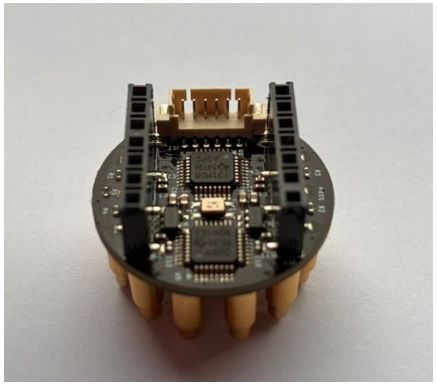
For research and education only. Not a medical device. Specifications subject to change.

02 FEATURE POSITIONS

Feature	Side	Position (mm, from centre)	Details
Board outline	—	centre 0, 0	Ø 26.00 circle, 1.6 mm thick
Pogo pins (18)	Pin side	see channel table	4 × 11 mm spring-loaded, SMD-mounted, hexagonal ≈ 5.38 pitch
REF pin	Pin side	+5.50, 0.00	Common reference (AINREF)
GND pin	Pin side	−5.49, 0.00	Ground
XIAO sockets (2 × 1 × 7)	Component side	rows at x = ±7.62, pins y = −7.62 … +7.62	2.54 mm pitch, 15.24 mm row spacing, SMD vertical
Stacking connector	Component side	0.00, +8.86	6-pin, 1.25 mm pitch, vertical
Converters (2)	Component side	0, +2.78 and 0, −8.01	5 × 5 mm packages (reference only)

Channel pin coordinates (pin-side view, +y toward the stacking connector)

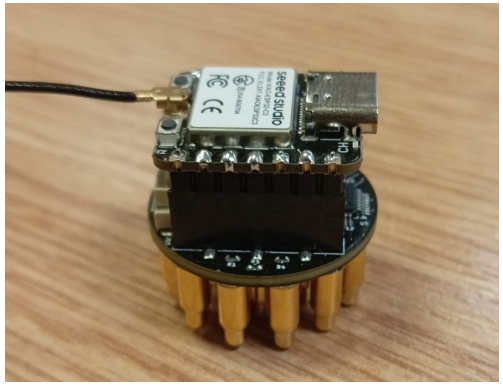
Ch	x	y	Ch	x	y	Ch	x	y	Ch	x	y
1	+10.14	-2.72	**5**	-2.72	-10.14	**9**	-10.14	+2.72	**13**	+2.72	+10.15
2	+7.43	-7.43	**6**	-2.75	-4.77	**10**	-7.43	+7.43	**14**	+2.75	+4.77
3	+2.75	-4.77	**7**	-7.42	-7.42	**11**	-2.75	+4.76	**15**	+7.43	+7.42
4	+2.72	-10.15	**8**	-10.14	-2.73	**12**	-2.72	+10.14	**16**	+10.14	+2.71



Component side with XIAO sockets.



Pin side: 18 pogo pins.



XIAO mounted with antenna.

NOTE Holder design.

Keep a clear Ø 27 mm opening for the pin cluster and press the board straight down. 3D holders: Thingiverse thing 7386761.

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